



Mathematics Part 1 — Cbse Annual Exam — 80 Marks

Subject: Mathematics Part 1 | Class: Class 9

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	To factor the expression $50p^2 + 60pq + 18q^2$ using the identity $(a+b)^2 = a^2 + 2ab + b^2$, what must first be taken out as a common factor? (A) 2 (B) 3 (C) 5 (D) 6 <i>[Ch 4: Exploring Algebraic Identities]</i>	1
2	Assertion (A): The distance between the points A (3, 4) and D (7, 1) is 5 units. Reason (R): The distance between two points is found by adding the difference in x-coordinates and the difference in y-coordinates. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	1
3	One diagonal of a rhombus is twice as long as the other. If the rhombus has area 128 cm ² , what is the length of the shorter diagonal? (A) $8\sqrt{2}$ cm (B) 16 cm (C) 12 cm (D) $4\sqrt{2}$ cm <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	1



Q.NO.	QUESTIONS	MARKS
4	Assertion (A): The side of a cube whose volume is $8n^3 - 60n^2m + 150nm^2 - 125m^3$ cubic units is $2n - 5m$ units. Reason (R): The expression $8n^3 - 60n^2m + 150nm^2 - 125m^3$ cannot be written in the form $(a-b)^3$ because the terms have negative coefficients. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 4: Exploring Algebraic Identities]</i>	1
5	A rational number in its lowest form has a denominator of $2^3 \times 5$. How many decimal places will its decimal expansion have? (A) 1 (B) 2 (C) 4 (D) 3 <i>[Ch 3: The World of Numbers]</i>	1
6	Assertion (A): $(x + y)(x^2 - xy + y^2) = x^3 + y^3$ is an identity. Reason (R): $(x - y)(x^2 + xy + y^2) = x^3 - y^3$ is an identity. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 4: Exploring Algebraic Identities]</i>	1
7	For the relationship $^{\circ}\text{C} = a \cdot ^{\circ}\text{F} + b$, where ice melts at 0°C and 32°F, and water boils at 100°C and 212°F, what is the value of b? (A) $-160/9$ (B) $160/9$ (C) $-32/9$ (D) $32/9$ <i>[Ch 2: Introduction to Linear Polynomials]</i>	1
8	Which mathematician was the first to solve algebraic problems using geometry by interpreting them in terms of coordinates in the plane? (A) Brahmagupta (B) Ömar Khayyām (C) Al-Bīrūnī (D) Fermat <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	1



Q.NO.	QUESTIONS	MARKS
9	To construct a line segment of length $\sqrt{2}$ on the number line, we measure OA = (A) 1 unit (B) $\sqrt{2}$ units (C) 2 units (D) $\sqrt{3}$ units <i>[Ch 3: The World of Numbers]</i>	1
10	On what date is Pi Approximation Day celebrated in India? (A) March 14 (B) March 22 (C) July 22 (D) July 14 <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	1
11	Lines with equal slopes but different y-intercepts are always: (A) Perpendicular (B) Intersecting (C) Coincident (D) Parallel <i>[Ch 2: Introduction to Linear Polynomials]</i>	1
12	In moving from point A(3, 4) to point D(7, 1), what is the distance covered along the x-axis? (A) 1 (B) 3 (C) 7 (D) 4 <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	1
13	For a square with side a, what is the ratio $P^2 : A$, where P is the perimeter and A is the area? (A) 16:1 (B) 4:1 (C) 8:1 (D) 2:1 <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	1



Q.NO.	QUESTIONS	MARKS
14	Why does $\sqrt{-1}$ not exist on the real number line? (A) Because there is no real number that, when multiplied by itself, results in a negative number. (B) Because the square root operation is only defined for positive integers. (C) Because $\sqrt{-1}$ is a rational number that cannot be plotted. (D) Because negative numbers were not formalized until the 19th century. <i>[Ch 3: The World of Numbers]</i>	1
15	What is the result of multiplying $(x + y + z)$ and $(x^2 + y^2 + z^2 - xy - xz - yz)$? (A) $x^3 + y^3 + z^3$ (B) $x^3 + y^3 + z^3 + 3xyz$ (C) $x^3 - y^3 - z^3 - 3xyz$ (D) $x^3 + y^3 + z^3 - 3xyz$ <i>[Ch 4: Exploring Algebraic Identities]</i>	1
16	Which of the following is a polynomial of degree 3 in the variable x with the coefficient of the x^2 term equal to -7? (A) $x^3 - 7x^2 + 2x + 1$ (B) $-7x^3 + 2x + 1$ (C) $3x^3 + 7x^2 - x + 5$ (D) $x^2 - 7x + 3$ <i>[Ch 2: Introduction to Linear Polynomials]</i>	1
17	When simplifying the rational expression $(x^2 - 7x + 12)/(5x^2 + 5x - 100)$, assuming the denominator is not zero, what is the final simplified form? (A) $(x - 4)/(5(x + 5))$ (B) $(x - 3)/(5(x + 5))$ (C) $(x - 3)/(x + 5)$ (D) $(x - 4)/(x + 5)$ <i>[Ch 4: Exploring Algebraic Identities]</i>	1
18	A letter is picked at random from the word 'PROBABILITY'. What is the theoretical probability of picking the letter B? (A) $1/11$ (B) $2/11$ (C) $1/2$ (D) $2/9$ <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	1



Q.NO.	QUESTIONS	MARKS
19	Assertion (A): The number of square tiles at Stage n of a growing pattern is given by $2n - 1$. Reason (R): The polynomial $2n - 1$ has degree 1, making it an example of a linear polynomial. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 2: Introduction to Linear Polynomials]</i>	1
20	Assertion (A): The equation $y = (9/5)(x - 273) + 32$ represents a linear relationship between the temperature in Kelvin and Fahrenheit. Reason (R): In the equation $y = ax + b$, if we change the values of b keeping the value of a fixed, the lines shift but remain parallel to the original line. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 2: Introduction to Linear Polynomials]</i>	1
Section B Attempt 5 out of 7.		
21	In a survey of 50 students, 15 students said they liked football. The number of students who like football is 15, and the ____ (frequency/relative frequency) is ____ (fill in the fraction or decimal). <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	2
	OR The height of water in a tank is given by $h(t) = 3 - 0.5t$. By what constant amount does the height decrease per month, and what does this represent? <i>[Ch 2: Introduction to Linear Polynomials]</i>	
22	How were the city streets of the Sindhu-Sarasvatī Civilisation constructed, and what distance apart were they? <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	2
	OR A die is rolled 50 times, and it lands on the number 4 exactly 8 times. Calculate the relative frequency of rolling a 4. <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	
23	Use a suitable identity to find the value of 43^2. <i>[Ch 4: Exploring Algebraic Identities]</i>	2
24	How did Archimedes express the area of a circle in terms of a right-angled triangle?	2



Q.NO.	QUESTIONS	MARKS
	<i>[Ch 6: Measuring Space: Perimeter and Area]</i>	
25	Show that $\frac{2}{3}$ and $\frac{4}{6}$ are equal rational numbers using the equality rule. <i>[Ch 3: The World of Numbers]</i>	2
Section C Attempt 5 out of 7.		
26	Find 5 rational numbers between $\frac{2}{5}$ and $\frac{3}{5}$. <i>[Ch 3: The World of Numbers]</i>	3
	OR In the coordinate representation (x, y), what do the variables x and y specifically represent regarding the position of a point P in 2-D space? <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	
27	Evaluate 65^2 using Śhrīdharāchārya's identity $a^2 = (a + b)(a - b) + b^2$. <i>[Ch 4: Exploring Algebraic Identities]</i>	3
	OR In a village fair, there are 3 popular snacks available: Samosa, Pakora, and Bhaji. For drinks, villagers can choose either Chai or Lassi. (i) List the sample space of all possible snack and drink combinations a person could choose at the fair. (ii) List the event 'Selecting Samosa as a snack.' <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	
28	A straight line is represented by the equation $y = ax + b$. At what point does this line cut the y-axis? If a line is given by $y = 3x - 2$, what is its y-intercept and what does it signify? <i>[Ch 2: Introduction to Linear Polynomials]</i>	3
29	A point lies in Quadrant II and another point lies in Quadrant IV of the Cartesian plane. State the signs of the x-coordinate and y-coordinate for points in each of these two quadrants. <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	3
30	When $d = 0$ in a 4-gon with sides a, b, c, d, show that the semi-perimeter of the 4-gon equals the semi-perimeter of the triangle with sides a, b, c. <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	3
Section D Attempt 4 out of 6.		
31	The sum of three numbers x, y, and z is 6, their product is 8, and the sum of their squares is 14. Using the identity $(x + y + z)(x^2 + y^2 + z^2 - xy - xz - yz) = x^3 + y^3 + z^3 - 3xyz$, find the value of $x^3 + y^3 + z^3$.	5



Q.NO.	QUESTIONS	MARKS
	<i>[Ch 4: Exploring Algebraic Identities]</i>	
	OR Points A(3, 4) and D(7, 1) are plotted in the Cartesian plane. Using the Baudhāyana-Pythagoras Theorem, find the distance AD step by step. Then state the general formula for the distance between any two points (x_1, y_1) and (x_2, y_2). Verify that the formula works even when coordinates are negative, using A'(-3, 4) and D'(-7, 1). <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	
32	A student has ₹500 in her savings bank account. She gets ₹150 every month as pocket money. Construct a table of values showing the total amount she will have at the end of the first 4 months. Find a linear expression to represent the amount she will have in the n^{th} month. Explain why this represents linear growth. <i>[Ch 2: Introduction to Linear Polynomials]</i>	5
	OR A basket contains 4 red balls and 5 blue balls. One ball is drawn and laid aside, and a second ball is drawn. Draw a tree diagram to represent the possible outcomes and probabilities. Use the tree diagram to calculate the probability of drawing a red ball and then a blue ball, and the probability of drawing 2 blue balls. <i>[Ch 7: The Mathematics of Maybe: Introduction to Probability]</i>	
33	Find the perimeter of a sector of a circle of radius 14 cm and sector angle 75°. Explain the steps involved in calculating the arc length and the total perimeter of the sector. (Use $\frac{22}{7}$ for π) <i>[Ch 6: Measuring Space: Perimeter and Area]</i>	5
34	Prove that $\sqrt{2}$ is an irrational number using the proof by contradiction method. Explain each logical step clearly and state why the initial assumption leads to a contradiction. <i>[Ch 3: The World of Numbers]</i>	5
Section E Attempt 3 out of 4.		



Q.NO.	QUESTIONS	MARKS
35	Case Study: Using Identities for Quick Calculations A mathematics club is exploring how algebraic identities simplify calculations. Rahul computes 43^2 by writing it as $(40 + 3)^2$ and using the identity $(a + b)^2 = a^2 + 2ab + b^2$. Priya calculates 104×96 by expressing it as $(100 + 4)(100 - 4)$ and applying $(x + y)(x - y) = x^2 - y^2$. The teacher explains that identities are equations true for all values of variables, unlike equations like $x^2 - 1 = 24$ which hold only for specific values such as $x = 5$ or $x = -5$. (i) Using the identity $(a + b)^2 = a^2 + 2ab + b^2$, find the value of 43^2 by writing it as $(40 + 3)^2$. (ii) Using the identity $(x + y)(x - y) = x^2 - y^2$, calculate 104×96. (iii) The expression $x^2 - 1 = 24$ is true only for $x = 5$ or $x = -5$. Is this expression an identity? Justify your answer. (iv) Using the expansion of $(a + b)^2$, determine when $(a + b)^2$ will be equal to $a^2 + b^2$. <i>[Ch 4: Exploring Algebraic Identities]</i>	5
	OR Case Study: City Map Coordinates A student is creating a 2-D map of a city center, inspired by how early Arabic maps used the Ujjayinī meridian as a zero-longitude reference. On the map, the main intersection is chosen as the point of intersection of the axes, called the origin. The horizontal road is the x-axis and the vertical road is the y-axis. A cafe is located 3 units to the right and 2 units upwards from the origin. A library is located 4 units to the left and 5 units upwards from the origin. A park is located 2 units to the left and 3 units downwards from the origin. The student marks these locations using coordinates. (i) What are the coordinates of the origin? (ii) The park is located 2 units to the left and 3 units downwards from the origin. In which quadrant does the park lie? (iii) The cafe is at (3, 2). A shop is located at (7, 2). Find the distance between the cafe and the shop. (iv) A bookstore is located on the y-axis, 6 units upwards from the origin, and a florist is located on the x-axis, 6 units to the right of the origin. Are the coordinates of the bookstore and the florist the same? Explain. <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	



Q.NO.	QUESTIONS	MARKS
36	Case Study: Mobile Network Plans and Linear Relationships Two mobile networks offer different monthly plans. Network A's monthly charge is given by the linear relationship $y = 2x + 10$, where x is the data used in GB and y is the total cost in ₹. Network B's plan is given by $y = 2x + 25$. Both networks have graphs that are straight lines. Because their equations have the same coefficient of x but different constant terms, their graphs are parallel lines. A third network offers a plan given by $y = 5x$. The constant term in each equation represents the y-intercept, which is the distance from the origin where the line cuts the y-axis. (i) What is the slope of the line representing Network A's plan? (ii) The graphs of Network A and Network B are parallel lines. What is the relationship between their slopes? (iii) The constant term in the equation $y = 2x + 25$ is the y-intercept. What does the y-intercept represent geometrically? (iv) The line representing the third network's plan, $y = 5x$, has no constant term ($b = 0$). Through which specific point does this line pass? <i>[Ch 2: Introduction to Linear Polynomials]</i>	5
37	Case Study: Room Layout with Coordinates Reiaan is planning his room layout on a coordinate plane. The room door extends from $D_1 (9.5, 0)$ to $R_1 (11.5, 0)$. The bathroom door extends from $B_1 (0, 1.5)$ to $B_2 (0, 4)$. A rectangular study table has three of its feet at the points $(8, 9)$, $(11, 9)$ and $(11, 7)$. Reiaan wants to verify the dimensions of each item and check whether the bathroom door is wider or narrower than the room door. He also needs to find the position of the fourth foot of the table to complete the layout. (i) What is the width of the room door? (ii) What are the coordinates of the fourth foot of the rectangular study table? (iii) Find the width of the bathroom door. Is the bathroom door narrower or wider than the room door? (iv) Both B_1 and B_2 have x-coordinate equal to 0. What does this indicate about the position of the bathroom door with respect to the y-axis? <i>[Ch 1: Orienting Yourself: The Use of Coordinates]</i>	5

